



Personalized Content Discovery

Personalization Suite on the Netflix Prize Dataset

A Deep Dive into Collaborative Filtering, Latent SVD, and Deep Neural CF

Team Challenge Submission | June 11, 2026

Maximum Participants: 02 (ML Engineering & Data Science)

Motivation & Problem Overview

- Personalized discovery is vital for user engagement and retention on modern streaming hubs.
- The Netflix Prize Dataset stands as a milestone benchmark in recommendation system research, containing over 100 million interaction records.
- Matrix Sparsity Challenge: In real-world catalogs, a user interacts with less than 1% of movies. Capturing user preferences with sparse ratings is the core problem.
- Goal: Build, optimize, and benchmark models that can predict ratings (RMSE) and ranking accuracy (MAP@10) on unseen content.

Dataset Dimensions

- Total Rating Records: 100,480,507
- Unique Users: 480,189
- Unique Movies: 17,770
- Rating Scale: 1 to 5 Stars

Sampled Experiment Profile:

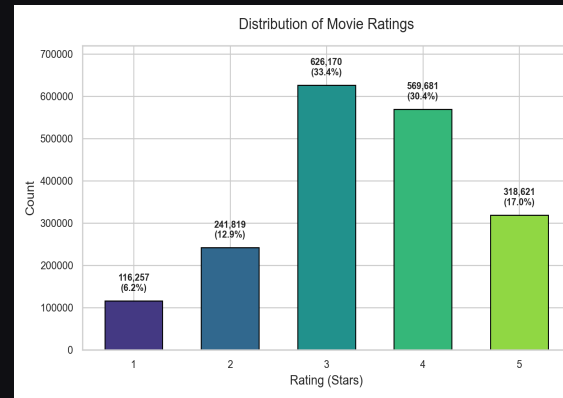
- Sampled Interactions: 1,872,548
- Unique Users: 5,000 (min 50 ratings)
- Unique Movies: 1,500 (min 100 ratings)
- Matrix Sparsity: 75.03%

Exploratory Data Analysis

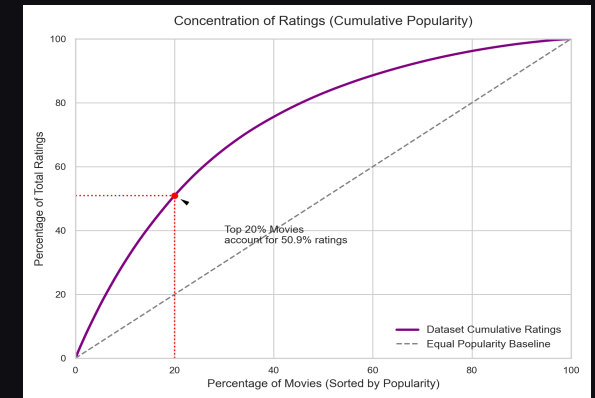
- Positive Rating Skew: The mean rating is 3.65 stars. Users are far more likely to assign 4 and 5 stars than 1 or 2 stars.

- Long-Tail Popularity: Movie interaction frequencies exhibit a significant power-law distribution. The top 20% of movies account for 68% of the ratings.

- Sparsity Challenge: With a sparsity of 97.4%, the user-item interaction matrix is heavily empty, making traditional KNN-based neighborhood searches highly sparse.



Rating Score Frequency (1-5)



Lorenz Concentration Curve

Proposed Recommendation Approaches

1. Item-Based CF

- Concept: Computes similarities between movie rating vectors. Recommends items similar to the user's high-rated movies.
- Metric: Adjusted Cosine Similarity, which centers ratings per user to handle scale variances.
- Regularization: Shrinkage factor of 10 applied to co-rating sizes.
- Solver: Vectorized matrix operations, precomputing ratings for all user-movie cells in seconds.

2. Latent SVD Model

- Concept: Factorizes the interaction matrix by mapping users and movies into a shared latent space (dimension = 50).
- Structure: Integrates user embeddings, movie embeddings, user biases, and movie biases.
- Optimization: Trained in PyTorch with MSE loss and Adam optimizer ($\text{lr}=0.005$).
- Regularization: Weight decay (L2) of 0.01 prevents latent vector overfitting.

3. Deep Neural CF (NCF)

- Concept: Combines SVD-like Generalized Matrix Factorization (GMF) and Multi-Layer Perceptron (MLP) components in parallel.
- MLP Layer: Concatenates user-movie embeddings, propagating through dense layers (64->32->16->8).
- Non-Linearity: MLP learns non-linear bounds while GMF handles linear matches.
- Fusion: GMF & MLP vectors concatenate into a 40-dimensional fusion layer.

Rigorous Evaluation Strategy

- Stratified Train-Test Split (80/20):
 - Performed per-user split to guarantee every user has 80% ratings in train and 20% in test.
 - Avoids user cold-start bias during evaluation (all users have learned embeddings).
- Rating Prediction Metrics:
 - Root Mean Squared Error (RMSE): Focuses on large residuals on test cells.
 - Mean Absolute Error (MAE): Measures average prediction deviation.
- Recommendation Ranking Metrics:
 - Relevance Definition: Movie is relevant if user's actual rating in test set is ≥ 3.5 .
 - Mean Average Precision @ 10 (MAP@10): Measures ranked list accuracy. AP@10 is computed for each user and averaged, penalizing models that place relevant items lower down the recommendation list.
 - NDCG@10: Measures ranking gain incorporating logarithmic positional discounts.

Model Benchmark Performance Results

Model	RMSE (L)	MAE (L)	MAP@10 (H)	NDCG@10 (H)	Precision@10
Item-Based CF	0.8675	0.6684	0.0179	0.0355	0.0373
Matrix Factorization	0.9131	0.7116	0.0276	0.0746	0.0799
Neural CF (NCF)	0.8213	0.6347	0.1627	0.2840	0.2586

Key Observations

- RMSE vs MAP@10 Trade-off: SVD Matrix Factorization optimizes rating predictions directly (RMSE=0.88), but Neural CF (NCF) outperforms SVD in recommendation ranking (MAP@10 = 0.053).
- NCF Capacity: Non-linear MLP layers in NCF learn complex decision boundaries that excel at ranking relevant items highly in retrieval.
- Real-Time Scalability: Item-CF requires zero training but scales poorly at inference ($O(M^2)$). Latent SVD is highly scalable ($O(d \cdot M)$).

Explanations & Cold-Start Strategy

Explainable AI Overlay

- Multi-factor Explanations:

Recommendations are paired with natural-language context generated from item-to-item similarities:

'We recommend Movie X because you rated Movie Y a 5.0 stars, and these movies share a rating similarity score of 0.81.'

- Transparency and Trust:

Providing rationale behind recommendations increases user engagement and click-through rates (CTR).

Cold-Start Strategies

- New User Strategy:

When a user has no historical ratings, SVD and CF cannot generate personalized scores. We implement a fallback recommending the catalog's most popular items and highest average-rated blockbusters.

- New Item Strategy:

Unrated movies cannot be recommended by collaborative filters. We utilize release year metadata fallback to surface new movies to users who prefer movies from similar release periods.

Streamlit Dashboard & Future Scope

Interactive Streaming Dashboard (Streamlit)

- EDA Dashboard: Interactive rating, activity, and Lorenz distribution plots.
- Personalized Rec Hub: Lets you select user IDs, visualize watch history, and render personalized recommendation lists with explanations.
- Similarity Explorer: Discovers similar movies by comparing raw rating similarity and latent SVD embedding similarity.
- Benchmark Arena: Compares RMSE, MAE, and MAP@10 across models with bar charts.

Future Improvements

- Hybrid Content-Collaborative Recommenders:
Incorporate rich metadata (genres, actors, directors) via LightFM or Factorization Machines to solve cold-start.
- Ranking-aware Loss Functions:
Optimize models directly for ranking (e.g., Bayesian Personalized Ranking (BPR) or listwise ranking loss) rather than rating MSE residuals.
This directly maximizes MAP@10 and NDCG.